# Sai Vinayaka Venkata Prateek Kacham

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## SUMMARY

AI/ML engineer with hands-on experience training deep learning models for perception and intelligent decision-making, fine-tuning large language models, and building multi-agent evaluation systems. Proficient in PyTorch, Hugging Face, and QLoRA. Pursuing MS in Information Systems at Northeastern University focused on Generative AI and Deep Learning. NPTEL-certified (Top 10%) in Python.

## EDUCATION

**Northeastern University, Boston, MA**
*Master of Science in Information Systems* *Expected May 2027*
GPA: 3.686/4.0

**Vasavi College of Engineering, Hyderabad, Telangana, India**
*Bachelor of Technology in Information Technology* *August 2024*

## SKILLS AND CERTIFICATIONS

**AI/ML:** PyTorch, TensorFlow, Hugging Face Transformers, PEFT, QLoRA, scikit-learn, LangChain, AutoGen, CNNs, GANs, NLP, RAG, NumPy, Matplotlib
**Programming:** Python, JavaScript, Java, C, SQL
**Web & Cloud:** React.js, Node.js, Express.js, MongoDB, AWS (EC2, S3), RESTful APIs, JWT
**Tools:** Git, Docker, Apache Kafka, Gradio, Google Colab (A100 GPU), Linux
**Certifications:** The Joy of Computing using Python – NPTEL (Top 10%)

## PROJECTS

**Mistral 7B Fine-Tuning for Text-to-SQL | Northeastern University — CSYE 7380**
*Python · PyTorch · Hugging Face · PEFT · QLoRA · BitsAndBytes · Gradio · A100 GP*

- Fine-tuned Mistral 7B Instruct v0.3 using QLoRA (4-bit quantization, LoRA r=16, alpha=32) on 5,000 samples, achieving +200% improvement in SQL exact match (0.11 → 0.33) over the base model
- Trained on Gretel synthetic text-to-SQL dataset using SFTTrainer for 3 epochs on A100 GPU; reduced training loss from 1.10 to 0.21
- Evaluated model performance using exact match, keyword F1, structural similarity, and ROUGE metrics across 100 unseen samples
- Deployed interactive Gradio demo enabling real-time natural language to SQL query generation

**Multi-Agent AI Pipeline for Perception & Validation | Northeastern University — CSYE 7380**
*Python · PyTorch · AutoGen · Apache Kafka · Google Colab*

- Trained CNN and ResNet-18 models on CIFAR-10 for image perception; designed an AutoGen Judge agent to autonomously evaluate and select the best-performing model based on confidence and accuracy metrics
- Implemented DCGAN with fully convolutional generator/discriminator (Conv2DTranspose + Conv2D) for image synthesis on CIFAR-10, demonstrating learned visual feature representations across 5,000 training epochs
- Developed NL-to-SQL translation pipeline using LLM prompt engineering (zero-shot and few-shot) to generate structured queries from natural language inputs
- Built Apache Kafka pub/sub system with 3 coordinating agents for real-time event-driven data streaming and automated processing

**Chatbot For Books Using Langchain**
*Python · LangChain · Groq LLaMA3 · ChromaDB · HuggingFace · Gradio · PyPDF2*

- Built a RAG-based intelligent chatbot using LangChain and Groq LLaMA3-8B for personalized book recommendations across multiple PDFs, serving 200+ monthly users
- Implemented ChromaDB vector store with HuggingFace embeddings (all-MiniLM-L6-v2) for semantic similarity search across chunked book content
- Tracked recommendation accuracy in real time via a live metrics panel — relevant response rate, total queries, and avg chunks retrieved
- Deployed interactive Gradio demo with multi-book comparison, conversation memory, and chat history export

**Flavour Fusion – Multi-Role Food Ordering Platform**
*React · Node.js · Express.js · MongoDB · JWT*

- Developed full-stack MERN application with role-based interfaces for customers, admins, restaurant owners, and delivery agents
- Architected Express.js REST API with MongoDB and JWT authentication, enabling seamless order tracking across all user roles

## RELEVANT COURSEWORK

- **Graduate Studies:** Generative AI & Deep Models, Application Engineering & Development, Program Structure & Algorithms, User Experience Design, Web Design/User Experience Engineering, Linux System Administrations